

ENERGY INFRASTRUCTURE ECONOMICS, POWER, FUELS AND ENERGY SECURITY

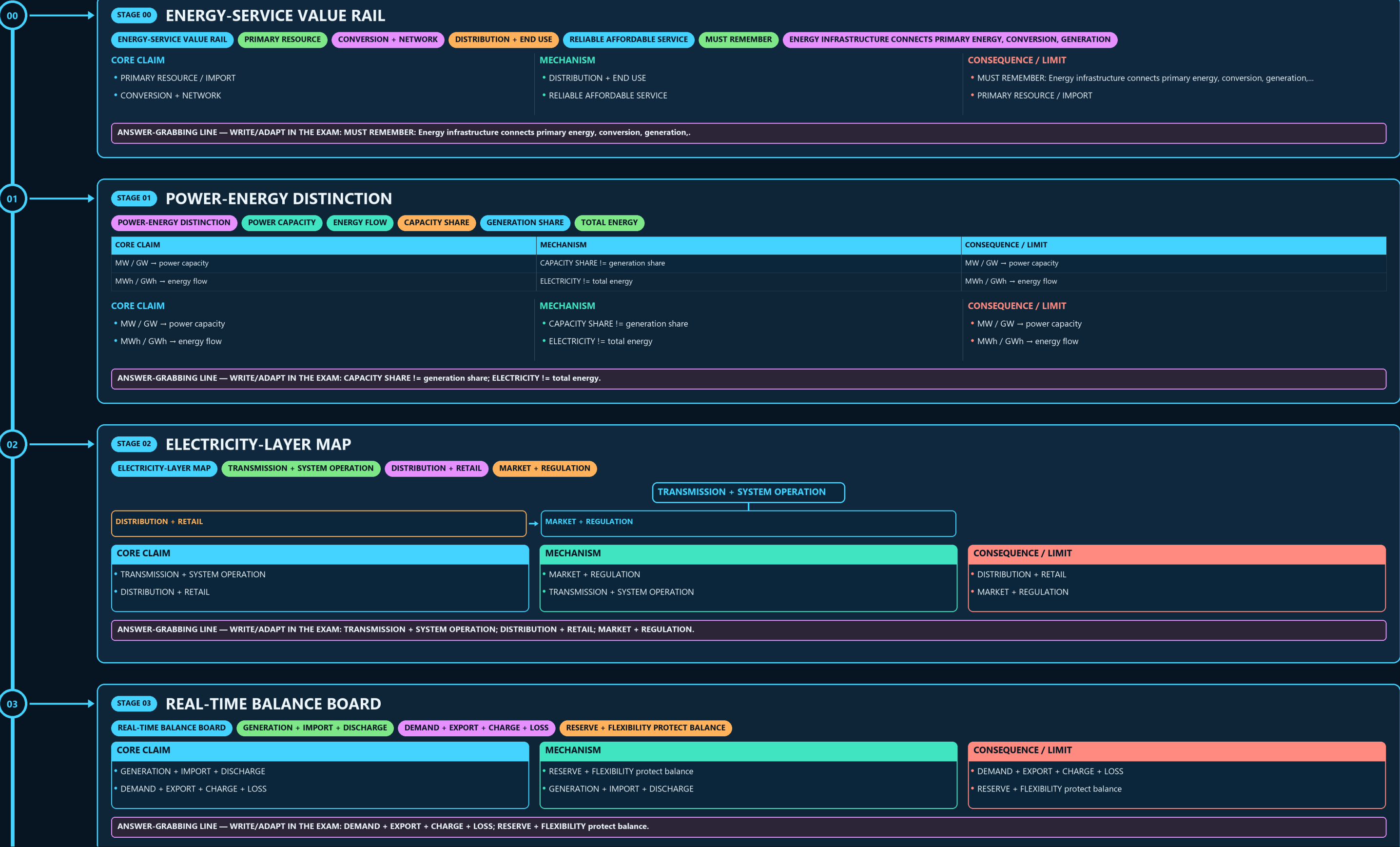
ENERGY-SERVICE VALUE RAIL → POWER-ENERGY DISTINCTION → ELECTRICITY-LAYER MAP → REAL-TIME BALANCE BOARD → INSTITUTIONAL PERIMETER → TARIFF-FINANCE SPLIT

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g3 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Capacity in MW is not generation in MWh, plant load factor is not.

06

STAGE 06 DISCOM DIAGNOSTIC

DISCOM DIAGNOSTIC ENERGY RECOVERY REVENUE-COST GAP PPA + SUBSIDY + COLLECTION ONE RATIO CANNOT DIAGNOSE ALL STRESS

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
AT&C → energy recovery	PPA + SUBSIDY + COLLECTION	AT&C → energy recovery
ACS-ARR → revenue-cost gap	ONE RATIO cannot diagnose all stress	ACS-ARR → revenue-cost gap
CORE CLAIM • AT&C → energy recovery • ACS-ARR → revenue-cost gap	MECHANISM • PPA + SUBSIDY + COLLECTION • ONE RATIO cannot diagnose all stress	CONSEQUENCE / LIMIT • AT&C → energy recovery • ACS-ARR → revenue-cost gap

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PPA + SUBSIDY + COLLECTION; ONE RATIO cannot diagnose all stress.

07

STAGE 07 PETROLEUM PERIMETER

PETROLEUM PERIMETER PIPELINE + STORAGE REFINING + MARKETING PNGRB EXCLUDES CRUDE + GAS PRODUCTION

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
• UPSTREAM → production	• DOWNSTREAM → refining + marketing	• UPSTREAM → production
• MIDSTREAM → pipeline + storage	• PNGRB excludes crude + gas production	• MIDSTREAM → pipeline + storage

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DOWNSTREAM → refining + marketing; PNGRB excludes crude + gas production.

08

STAGE 08 ENERGY-SECURITY DIMENSIONS AND RESILIENCE PENTAGON

ENERGY-SECURITY DIMENSIONS AND RESILIENCE PENTAGON PHYSICAL AVAILABILITY INFRASTRUCTURE ACCESSIBILITY USER AFFORDABILITY SOCIAL ACCEPTABILITY + SYSTEM RESILIENCE

PHYSICAL AVAILABILITY		
INFRASTRUCTURE ACCESSIBILITY →	USER AFFORDABILITY →	SOCIAL ACCEPTABILITY + SYSTEM RESILIENCE
CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
• PHYSICAL AVAILABILITY	• USER AFFORDABILITY	• PHYSICAL AVAILABILITY
• INFRASTRUCTURE ACCESSIBILITY	• SOCIAL ACCEPTABILITY + SYSTEM RESILIENCE	• INFRASTRUCTURE ACCESSIBILITY

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: INFRASTRUCTURE ACCESSIBILITY; USER AFFORDABILITY; SOCIAL ACCEPTABILITY + SYSTEM RESILIENCE.

09

STAGE 09 RENEWABLE-INTEGRATION CHAIN

RENEWABLE-INTEGRATION CHAIN VARIABLE CAPACITY GRID + FORECAST FLEXIBILITY + STORAGE RELIABLE GENERATION VALUE

VARIABLE CAPACITY		
GRID + FORECAST →	FLEXIBILITY + STORAGE →	RELIABLE GENERATION VALUE
CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
• VARIABLE CAPACITY	• FLEXIBILITY + STORAGE	• VARIABLE CAPACITY
• GRID + FORECAST	• RELIABLE GENERATION VALUE	• GRID + FORECAST

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: GRID + FORECAST; FLEXIBILITY + STORAGE; RELIABLE GENERATION VALUE.

09

STAGE 09 RENEWABLE-INTEGRATION CHAIN

RENEWABLE-INTEGRATION CHAIN VARIABLE CAPACITY GRID + FORECAST FLEXIBILITY + STORAGE RELIABLE GENERATION VALUE

GRID + FORECAST

FLEXIBILITY + STORAGE

VARIABLE CAPACITY

RELIABLE GENERATION VALUE

CORE CLAIM

- VARIABLE CAPACITY
- GRID + FORECAST

MECHANISM

- FLEXIBILITY + STORAGE
- RELIABLE GENERATION VALUE

CONSEQUENCE / LIMIT

- VARIABLE CAPACITY
- GRID + FORECAST

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: GRID + FORECAST; FLEXIBILITY + STORAGE; RELIABLE GENERATION VALUE.

10

STAGE 10 STATUS AND UNIT LADDER

STATUS AND UNIT LADDER INSTALLED CAPACITY GENERATION + SERVICE OUTCOME ANNOUNCED / BILL

APPROVED / SANCTIONED / OUTLAY

INSTALLED CAPACITY

ANNOUNCED / BILL

GENERATION + SERVICE OUTCOME

CORE CLAIM

- ANNOUNCED / BILL
- APPROVED / SANCTIONED / OUTLAY

MECHANISM

- INSTALLED CAPACITY
- GENERATION + SERVICE OUTCOME

CONSEQUENCE / LIMIT

- ANNOUNCED / BILL
- APPROVED / SANCTIONED / OUTLAY

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: APPROVED / SANCTIONED / OUTLAY; INSTALLED CAPACITY; GENERATION + SERVICE OUTCOME.

11

STAGE 11 ENERGY-ECONOMICS ANSWER SPINE

ENERGY-ECONOMICS ANSWER SPINE DEFINE CARRIER + UNIT + LAYER TRACE NETWORK + TARIFF + REGULATOR TEST SECURITY + TRANSITION CONCLUDE SERVICE, NOT HEADLINE CAPACITY EVIDENCE LIMIT STATE FUEL/TECHNOLOGY, STOCK OR FLOW, UNIT

DEFINE carrier + unit + layer

TRACE network + tariff + regulator

TEST security + transition

CONCLUDE service, not headline capacity

EVIDENCE LIMIT: FORMULA / STATUS / CAUSATION: State fuel/technology, stock or flow, unit

CORE CLAIM

- DEFINE carrier + unit + layer
- TRACE network + tariff + regulator

MECHANISM

- TEST security + transition
- CONCLUDE service, not headline capacity

CONSEQUENCE / LIMIT

- EVIDENCE LIMIT: FORMULA / STATUS / CAUSATION: State fuel/technology, stock or flow, unit,...
- DEFINE carrier + unit + layer

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: FORMULA / STATUS / CAUSATION: State fuel/technology, stock or flow, unit,.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT DISTRIBUTIONAL AND TRANSITION COST OVERCONSUMPTION IF VARIABLE PRICE IS TOO LOW EXCLUSION IF FIXED CHARGE IS TOO HIGH REVENUE INSTABILITY IF FIXED NETWORK COST IS RECOVERED MAINLY SHIFTS BURDEN TO FUTURE CONSUMERS

CAPACITY NO LONGER NEEDED OLD HIGH TARIFFS AFTER TECHNOLOGY COST FALLS

OPTIONAL SOURCE DEPTH

- distributional and transition cost
- overconsumption if variable price is too low;
- exclusion if fixed charge is too high;
- revenue instability if fixed network cost is recovered mainly per kWh while rooftop/self

ADVANCED DEBATE

- shifts burden to future consumers;
- capacity no longer needed;
- old high tariffs after technology cost falls;
- fuel/transport assumptions that later fail.

LEGACY / NUANCE

- performance and availability incentives;
- transmission readiness before commissioning.
- suppress wholesale energy prices;
- reduce thermal running hours;

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.